



### 3.22 AVIATION TECHNOLOGY (450)

#### 3.22.1 Aviation Technology Paper 1 (450/1)

##### SECTION A (44 marks)

*Answer **all** questions in this section.*

- 1 (a) List **four** causes of electric shock as laid down in safety practices. (2 marks)  
(b) State **one** reason for bonding an aircraft during refuelling. (1 mark)
- 2 (a) Differentiate between the following terms as applied to materials:  
(i) Mechanical properties; (1 mark)  
(ii) Chemical properties. (1 mark)  
(b) Sketch each of the following aircraft hardware and state where each is applied:  
(i) Spring washer; (1 mark)  
(ii) Woodruff key; (1 mark)  
(iii) Countersunk rivet. (1 mark)
- 3 (a) Outline **two** environmental issues associated with the aviation industry. (2 marks)  
(b) Name **two** types of clouds with vertical development. (1 mark)
- 4 (a) List **three** challenges likely to affect the performance of a well managed international airline. (3 marks)  
(b) State the functions of each of the following in a Non-Destructive testing process:  
(i) Penetrants; (1 mark)  
(ii) Developers. (1 mark)
- 5 State the methods used to join each of the following aircraft parts:  
(a) Honey-comb structures; (1 mark)  
(b) Engine fire wall. (1 mark)
- 6 Explain the principle of generating thrust in each of the following aircraft engines:  
(a) reciprocating engine; ( $2\frac{1}{2}$  marks)



- (b) pure jet engine. (2  $\frac{1}{2}$  marks)

7 Name **five** stresses acting on an aircraft in flight and state where each is applicable. (5 marks)

**Types of stress**

**Where applicable**

8 (a) Outline **four** functions of an aircraft undercarriage. (4 marks)

(b) State **three** basic principles of air navigation. (3 marks)

9 Outline **five** methods of reducing skin-friction associated with the boundary layer in flight. (5 marks)

10 Sketch and show the abbreviation of the following as applied in engineering drawing:

(a) across flats; (1 mark)

(b) under cut; (1 mark)

(c) counterbore; (1 mark)

(d) centreline. (1 mark)

### SECTION B (56 marks)

*Answer any **four** questions from this section.*

11 (a) List **four** causes of 75% of air accidents caused by human error in the aviation industry. (2 marks)

(b) Explain **three** business opportunities for chartered aircraft operators. (6 marks).

(c) Explain the role of each of the following in the Aviation Industry:

(i) marshaller; (1  $\frac{1}{2}$  marks)

(ii) flight Engineer; (1  $\frac{1}{2}$  marks)

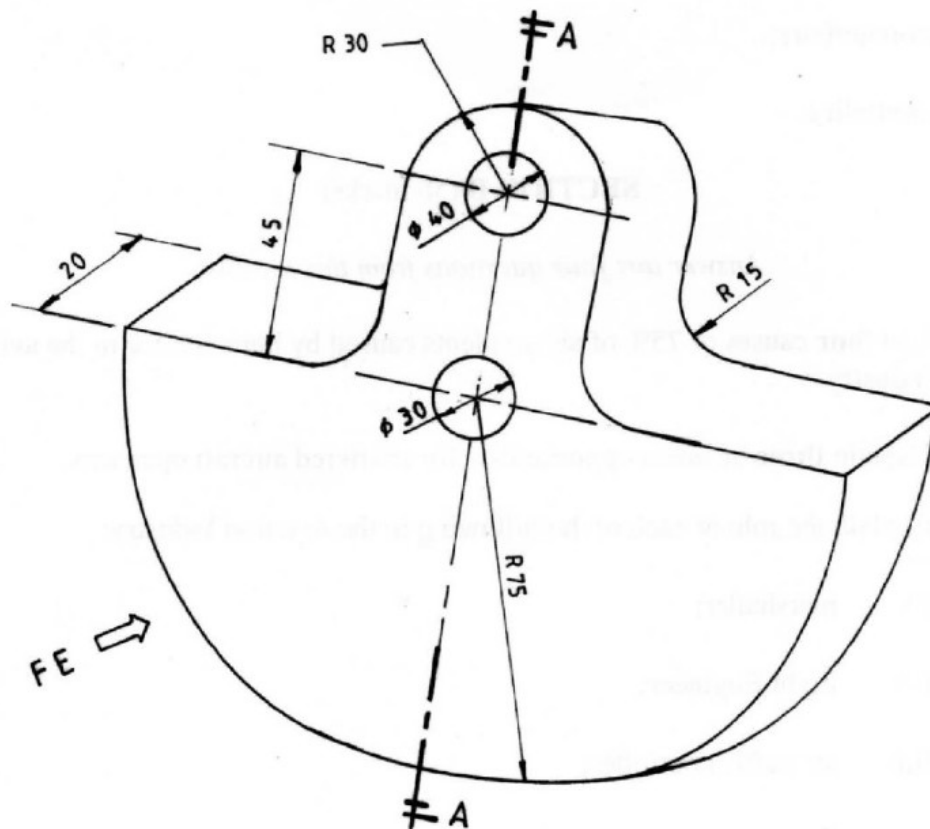
(iii) air traffic controller; (1  $\frac{1}{2}$  marks)

(iv) Purser. (1  $\frac{1}{2}$  marks)



- 12 (a) Outline **four** methods used to vary the flow of fuel in an aircraft engine. (4 marks)
- (b) Explain the principle of operation of an aero piston engine float type carburettor. (4 marks)
- (c) Explain the principle of operation of each of the following aircraft gas turbine engine burners:
- (i) simplex burner; (3 marks)
  - (ii) duplex burner. (3 marks)
- 13 (a) Explain the meaning of **four** colour markings on aircraft instruments. (4 marks)
- (b) Define the term graticule as applied in navigation. (1 mark)
- (c) Aircraft A is flown from point X,  $26^{\circ} 30' N$  to point Y,  $44^{\circ} 30' N$  while aircraft B is flown from point Q  $170^{\circ} E$  to point P,  $165^{\circ} W$ . Determine the change in latitude and longitude respectively for aircrafts A and B in degrees and minutes. (9 marks)

14 **Figure 1** shows an aircraft hinge bracket, drawn in isometric projection.





Draw FULL SIZE in first angle projection the following views:

- (a) front elevation in the direction of arrow F.E.
- (b) Section A-A.
- (c) Plan

(Use the A3 paper provided)

(14 marks)

- 15 (a) Describe each of the following aircraft structural members:

- (i) bulk head; (2 marks)
- (ii) longerons; (2 marks)
- (iii) spars. (2 marks)

- (b) **Figure 2** shows a basic hydraulic system. Identify the components labelled 1 to 6.

(3 marks)

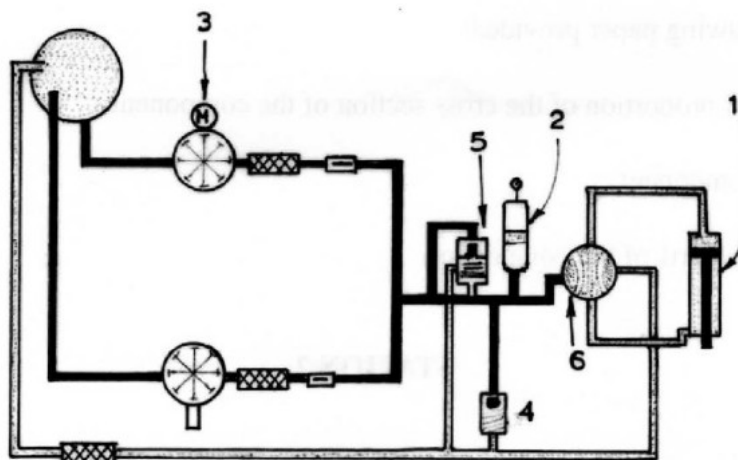


Figure 2

- 1. ....
- 2. ....
- 3. ....
- 4. ....
- 5. ....
- 6. ....

- (c) Outline **five** reasons why a thread cutting tap may break while in use.

(5 marks)



### 3.22.2 Aviation Technology Paper 2 (450/2)

#### STATION 1

##### INSTRUCTIONS

Figure 1 shows a pictorial drawing of a hydraulic system component.

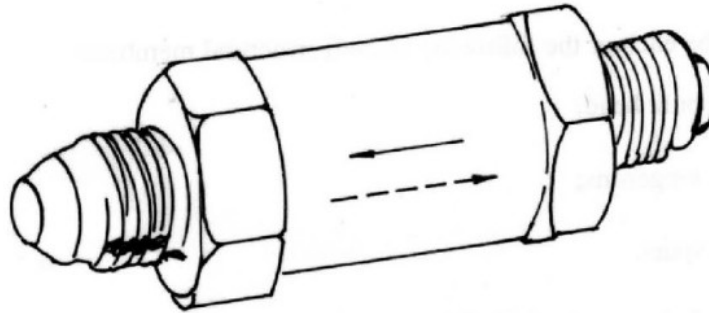


Figure 1

On the A3, drawing paper provided:

- (a) Sketch in good proportion of the cross section of the component.
- (b) Identify the component.
- (c) Label **six** main parts of the component.

(10 marks)

#### STATION 2

##### INSTRUCTIONS

Using the template, tools, and materials provided, make the aircraft ram air scoop shown in figure 2. (10 marks)

#### STATION 3

##### INSTRUCTIONS

Using the tools and materials provided, carry out the following tasks:

- (a) (i) Scribe each of the materials labeled A, B and C. Record your observations. (1  $\frac{1}{2}$  marks)
- (ii) Centre punch each material in a(i). Record your observations. (1  $\frac{1}{2}$  marks)
- (iii) File each of the materials in a(i). Record your observations. (1  $\frac{1}{2}$  marks)



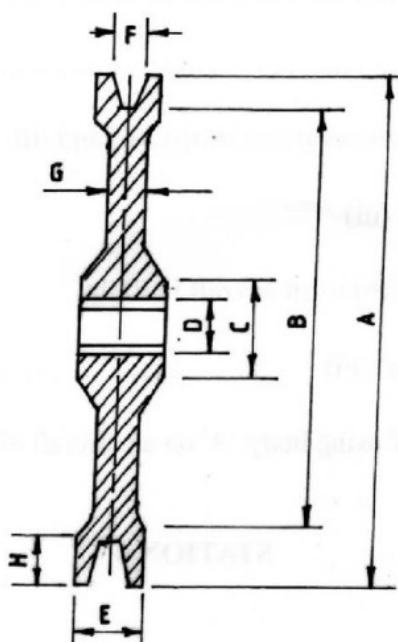


- (iv) Bend each of the materials in a(i) to 90°. Record your observations. (1  $\frac{1}{2}$  marks)
- (b) From the observations (i) - (iv), comment on the hardness of each material. (1  $\frac{1}{2}$  marks)
- (c) Identify the materials labelled A and C. (1 mark)
- (d) State where each of the materials in a(i) can be used on an aircraft. (1  $\frac{1}{2}$  marks)

#### SECTION 4

#### INSTRUCTIONS

- (a) Figure 3 shows a sectional view of an aircraft pulley provided.



- Using the tools, measure and record the dimensions A- H as shown in the figure. (8 marks)
- (b) Identify the function of the part painted red on the pulley. ( $\frac{1}{2}$  mark)
- (c) State **two** rejection criteria and **one** maintenance aspect of the pulley.
- (i) Rejection criteria ..... (1 mark)
- (ii) Maintenance aspect ..... ( $\frac{1}{2}$  mark)



## STATION 5

### INSTRUCTIONS

Study the set up provided and perform the following tasks:

- (a) (i) Pour 300 ml of water at point A. Record the amount collected in beaker A1.  
Amount of water .....
- (ii) Repeat a(i) at B. Record the amount of water collected in beaker B1.  
Amount of water .....
- (iii) Repeat a(i) at C. Record the amount of water collected in beaker C1.  
Amount of water ..... (6 marks)
- (b) State the reasons for your observations in a(i), (ii) and (iii)  
Observation in (i), (ii) and (iii) ..... ( $1\frac{1}{2}$  marks)
- (c) Relate your observation in (b) to an aircraft in flight.  
Relationship in b(i), (ii) and (iii) ..... ( $1\frac{1}{2}$  marks)
- (d) State **two** overall effects of using body 'A' on an aircraft effects ..... (1 mark)

## STATION 6

### INSTRUCTIONS

Using the tools and materials provided, carry out the following tasks.

- (a) Identify the materials labelled 'X' and 'Y' ..... (1 mark)
- (b) Strip one end of material labelled 'X' and name the parts. .... (2 marks)
- (c) (i) Tin the stripped end of material labelled 'X'.  
(ii) Clip the male and female connectors on the material labelled 'Y'. .... (7 marks)



## STATION 7

### INSTRUCTIONS

Using the tools, wheel assembly and brake assembly provided, carry out the following

- (a) On the whole assembly
- (i) Check the tyre pressure.
- Pressure ..... (1 mark)
- (ii) Torque load the nut painted red to 60 450 lbs . Let the examiner check your work. (1 mark)
- (iii) Identify the tyre size and ply rating.
- Size .....
- Ply rating ..... (1 mark)
- (iv) Identify and record **two** rejection criteria.
- Criteria (i) ..... (1 mark)
- (v) State the purpose of the blue and white marks.
- Blue .....
- White ..... (1 mark)
- (b) Safely wire the bolts on the brake assembly. (5 marks)

## STATION 8

### INSTRUCTIONS

Using the tools and the engine components provided, carry out the following tasks:

- (a) (i) Identify the components labelled M and N.
- (ii) Measure and record the gap of the component labelled N.
- Gap ..... (2 marks)
- (b) (i) Remove the cap on the component M.
- Let the examiner check your work.





- (ii) Identify the parts painted white and yellow.

White .....

Yellow .....

- (iii) State the condition of distribution points.

- (iv) Replace the cap.

Let the examiner check your work.

(4 marks)

- (c) (i) Fit the leads on the component marked M to match the best firing order.

Let the examiner check your work.

- (ii) State **two** rejection criteria of the part labelled P.

- (iii) State **two** functions of the part painted blue on component M.

(4 marks)

### STATION 9

#### INSTRUCTIONS

Study aircraft instruments marked 1 to 9 and carry out the following tasks.

(6 marks)

- (a) Group the instruments into three major categories and complete the table below.

| GROUP NAME | RELATED INSTRUMENTS  |
|------------|----------------------|
| 1.         | (i)<br>(ii)<br>(iii) |
| 2.         | (i)<br>(ii)<br>(iii) |
| 3.         | (i)<br>(ii)<br>(iii) |

- (b) State the reasons for each grouping in (a).

GROUP      REASON

(1½ marks)



(c) Name **three** settings made on instrument labelled 10. (1  $\frac{1}{2}$  marks)

(d) Name **one** common error and maintenance task for instruments labelled 1 to 9. (1 mark)

Error .....

Maintenance task .....

### STATION 10

#### INSTRUCTIONS

(a) Demonstrate to the examiner six marshalling signals from starting to taxing out on a twinned engine aircraft. (3 marks)

(b) Identify and state the use of areas labelled A, B, C, D and E. (5 marks)

AREA NAME

USE

(c) (i) State the name of runway and the direction of take off/landing.

Runway .....

Direction .....

(ii) Give the functions of G and H. (2 marks)

G .....

H .....